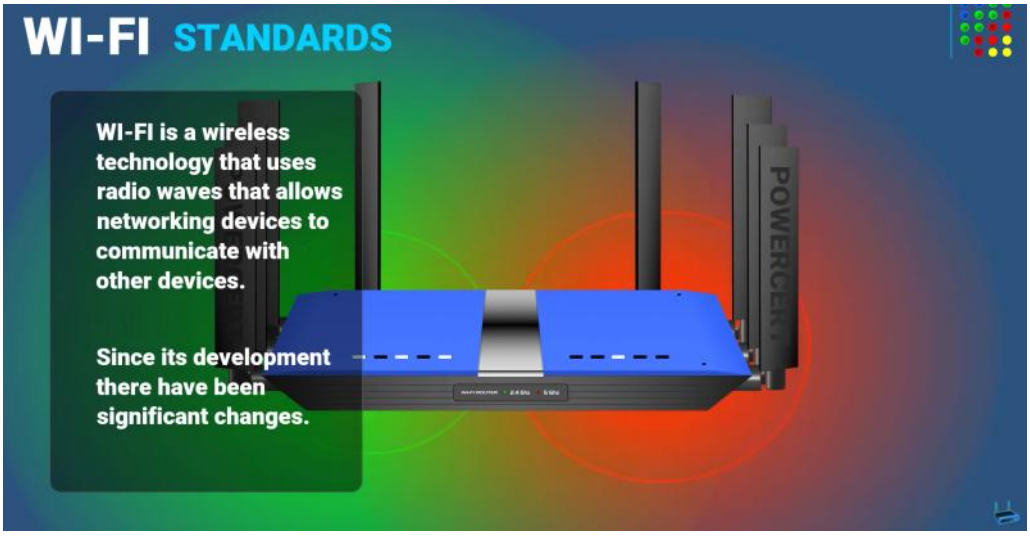
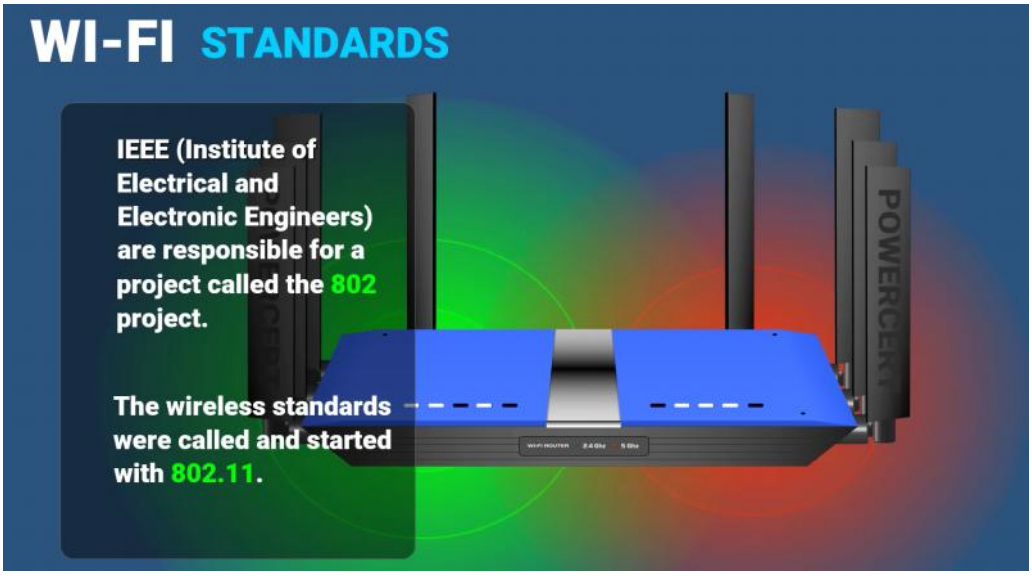
Wi-Fi Standards

Thursday, March 19, 2026 1:22 PM

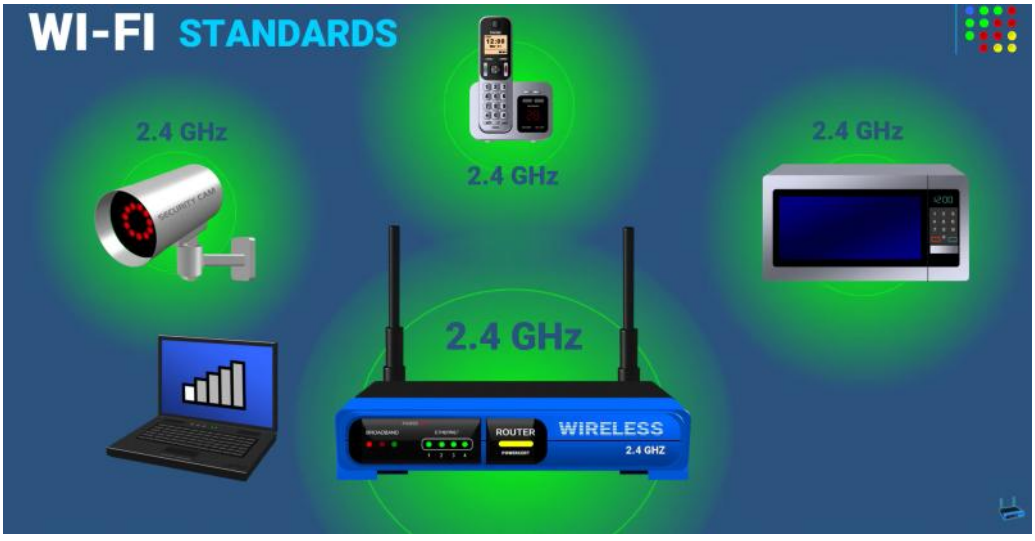
WiFi has evolved from very slow speeds around 1 to 2 Mbps in early days to extremely high speeds up to 46 Gbps with WiFi 7. Early standards used only 2.4 GHz which had more range but more interference. Later 5 GHz and 6 GHz bands were added to provide higher speeds and less congestion.

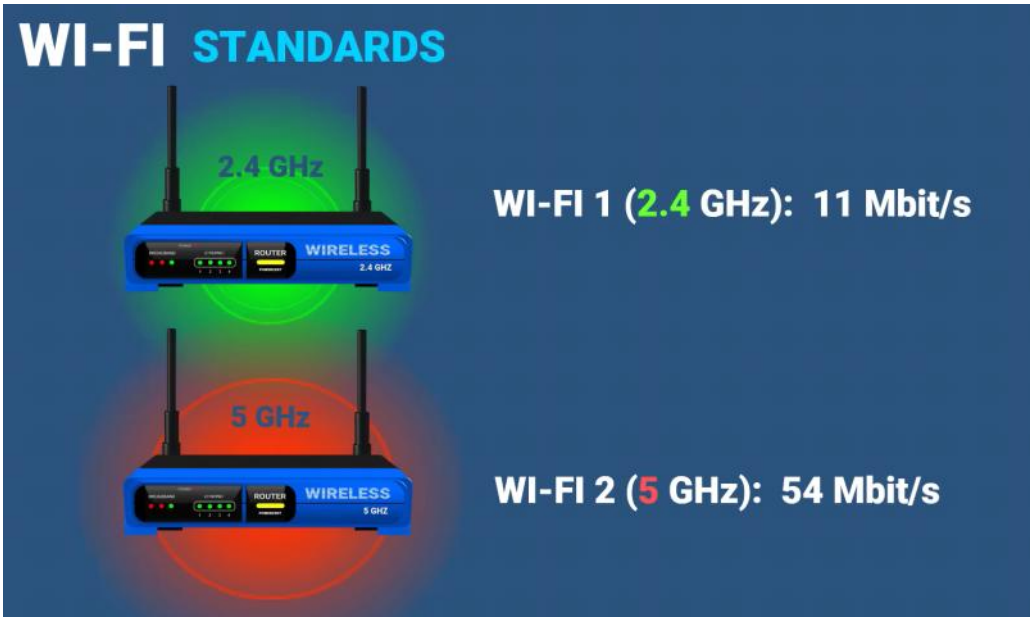


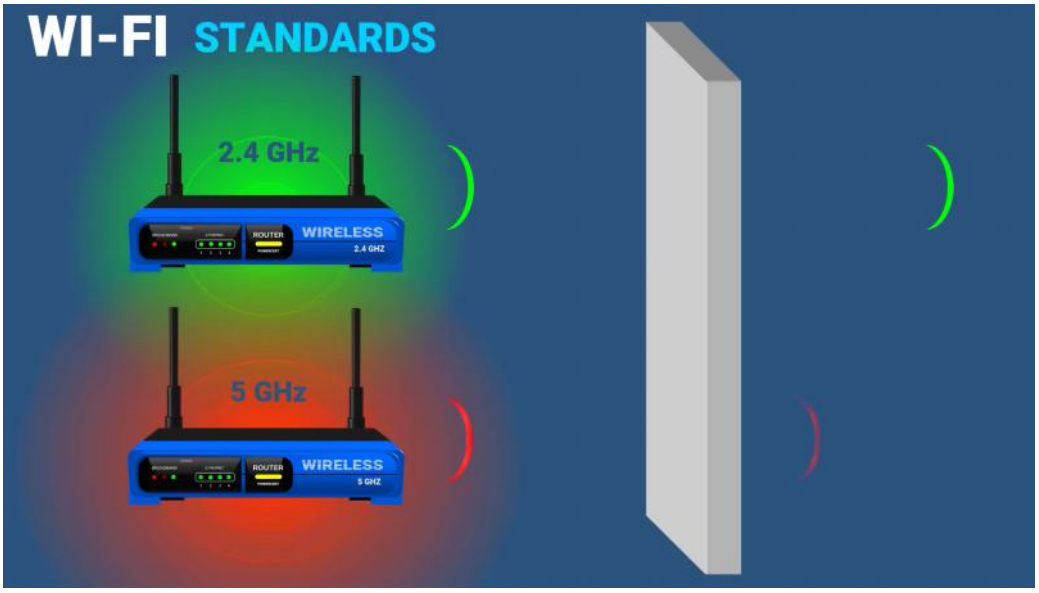


WI-FI STANDARDS
802.11b was the next wireless standard.
Also known as WI-FI 1
Came out in 1999.
Speed was up to 11 Mbit/s
Used the 2.4 GHz radio frequency band.
WI-FI 1
ROUTER
WIRELESS
2.4 GHZ

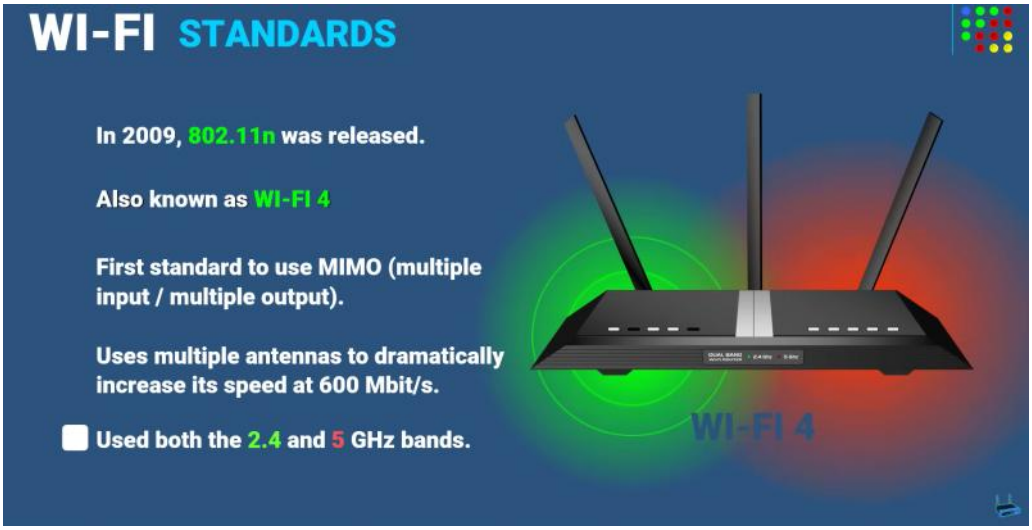
WI-FI STANDARDS
802.11a was the next wireless standard.
Came out in 1999.
Also known as WI-FI 2
The first standard to use the
5 GHz frequency band.
The difference between the 2.4 and 5 GHz
are interference, speed, and distance.
WI-FI 2
ROUTER
WIRELESS
5 GHZ

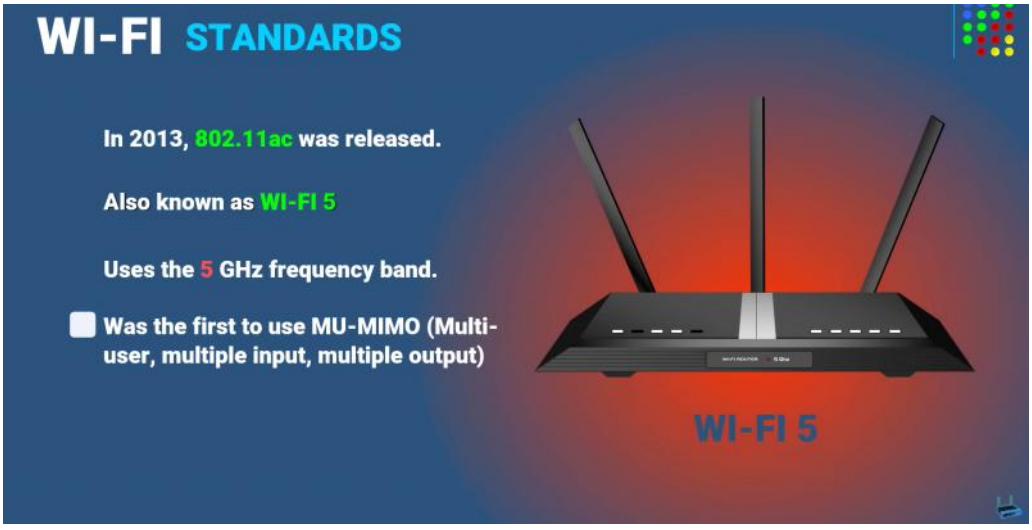

WI-FI STANDARDS
2.4 GHz
2.4 GHz
2.4 GHz
2.4 GHz
ROUTER
WIRELESS
2.4 GHZ



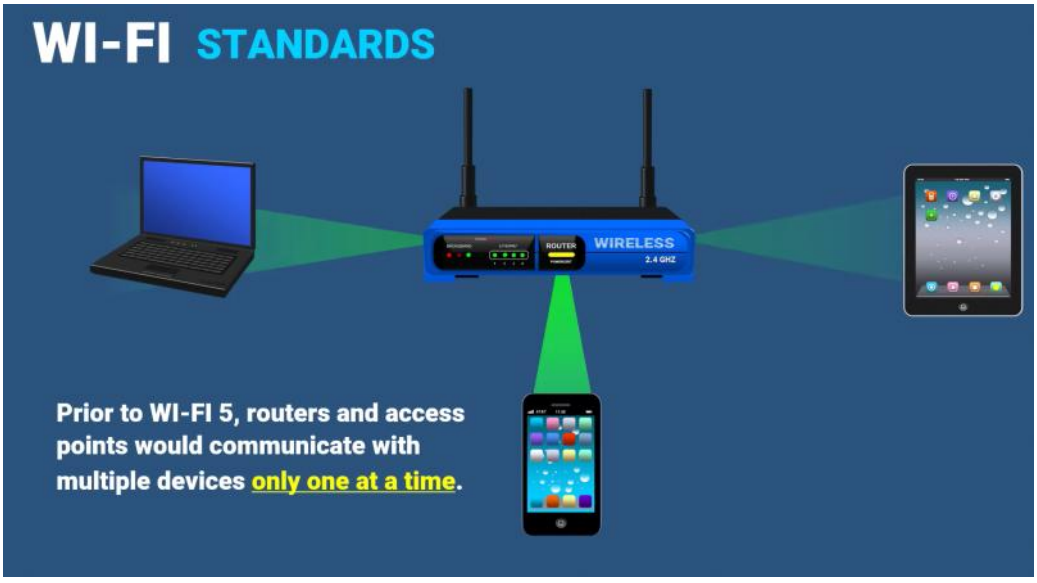


Wifi 5 GHz is more faster but has less range as it uses high frequency as it is difficult to penetrate through solid objects like floors and walls on buildings





Major improvements came with technologies like MIMO which uses multiple antennas, MU MIMO which supports multiple devices at once, and OFDMA which improves performance in crowded networks. Beamforming helps send signals directly to devices for better connection.


WI-FI STANDARDS
ROUTER
WIRELESS
2.4 GHZ
Prior to WI-FI 5, routers and access points would communicate with multiple devices only one at a time.

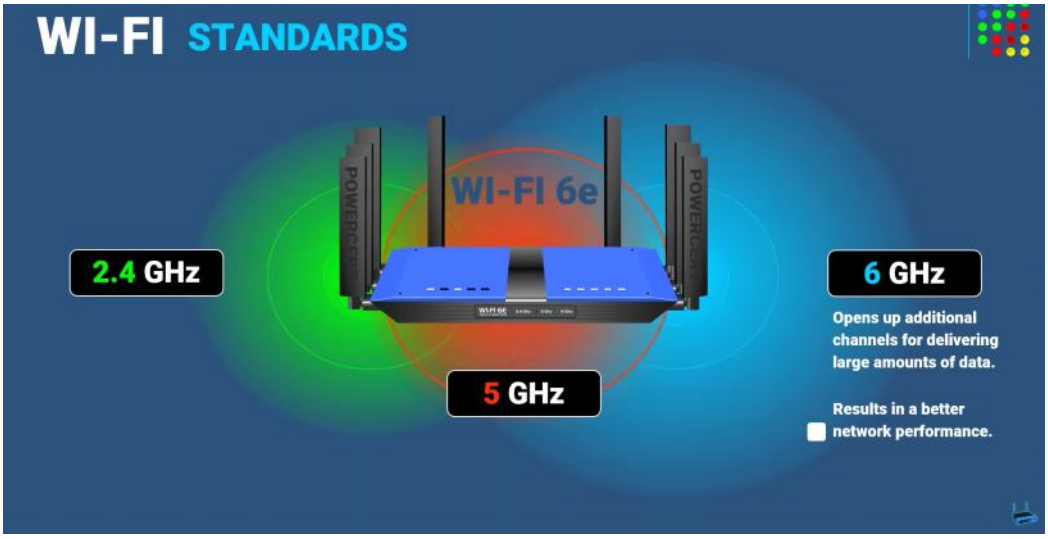
WI-FI STANDARDS
WI-FI 5 MU-MIMO communicates with multiple devices at the same time.
ROUTER
WIRELESS
2.4 GHZ
SMART TV

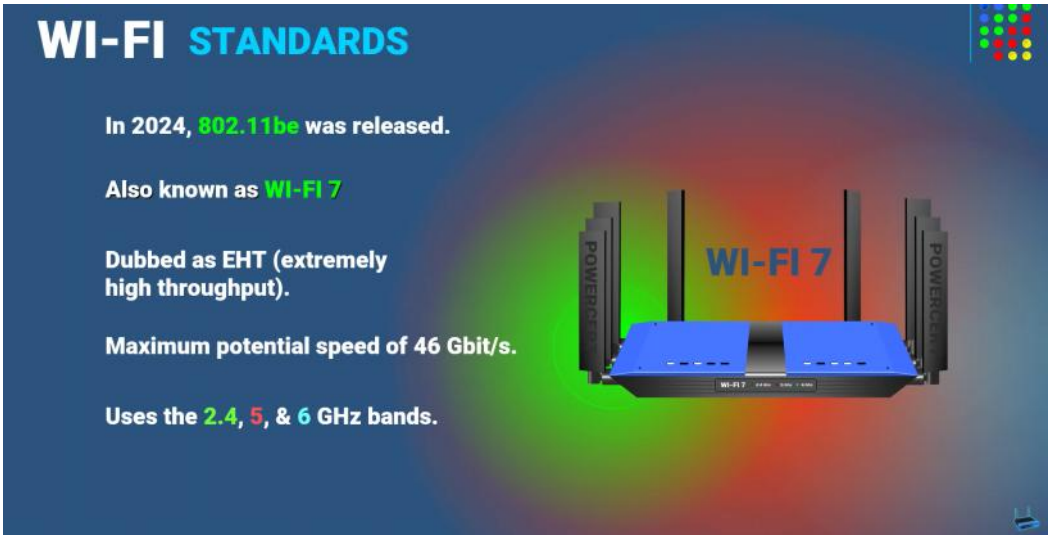
WI-FI STANDARDS
ROUTER
WIRELESS
2.4 GHZ
Beamforming
Without Beamforming

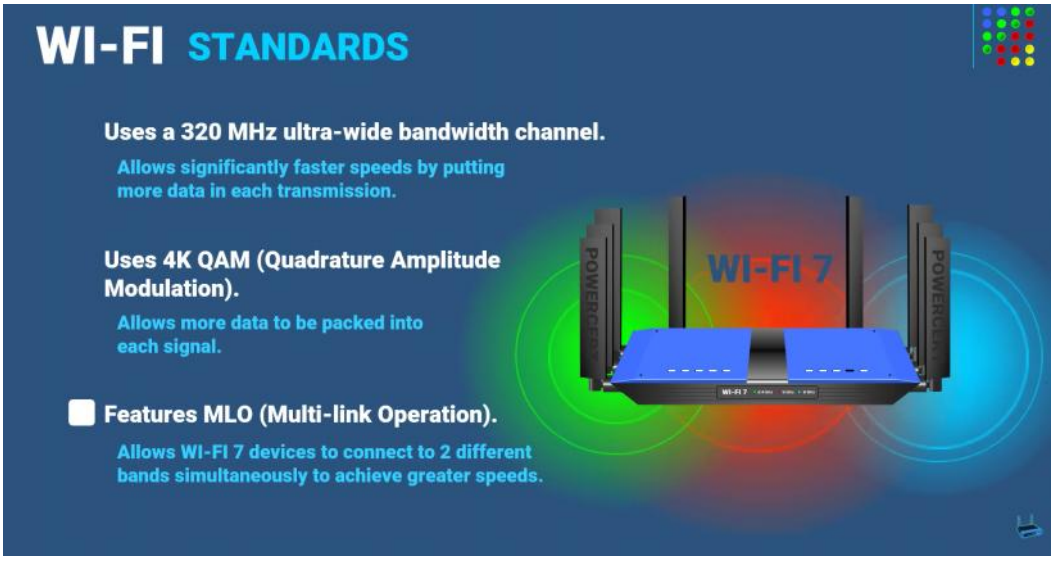
WI-FI STANDARDS
In 2019, 802.11ax was released.
Also known as WI-FI 6
Uses both the 2.4 and 5 GHz bands.
Maximum speed is 9.6 Gbit/s (shared across all devices).
The main goal of WI-FI 6 was to make a WI-FI network perform better when a lot of devices are connected to it.
WI-FI 6

WI-FI STANDARDS
WI-FI 6
OFDMA (Orthogonal frequency-division multiple access).

WI-FI STANDARDS
SMART TV
WI-FI 6
Improvement in MU-MIMO can support up to 12 simultaneous streams.







WiFi 7 brings major upgrades like wider channels, higher data capacity, and the ability to use multiple bands together for faster and more stable connections. Future WiFi 8 focuses more on reliability instead of just speed.

WiFi 8 is due to be released in 2028 and is under development.

Overall WiFi evolution focuses on higher speed better efficiency and improved performance for many connected devices.